

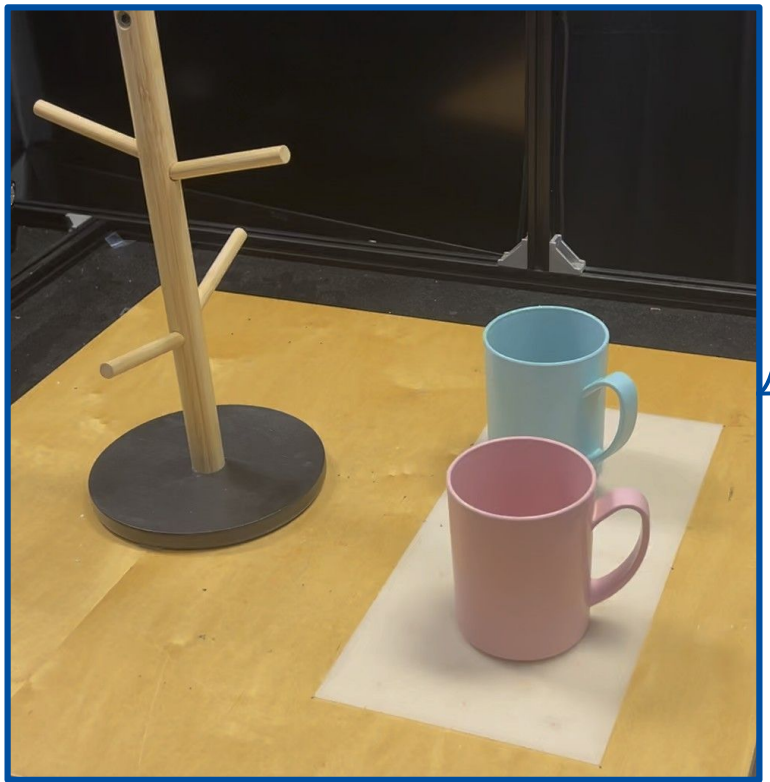
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Background

Via CodeDiffuser:

“Hang the blue mug
on a branch”

Vision Language Model



```
# find a mug
mugs = detect("mug")
mug = sel_name(mugs,"blue
mug")

# find a branch
branches = detect("branch")
branch =
rand_choice(branches)
```

 Can we do better than random choice?

Methods

"put a mug on a branch"

Generate code by LMP

```

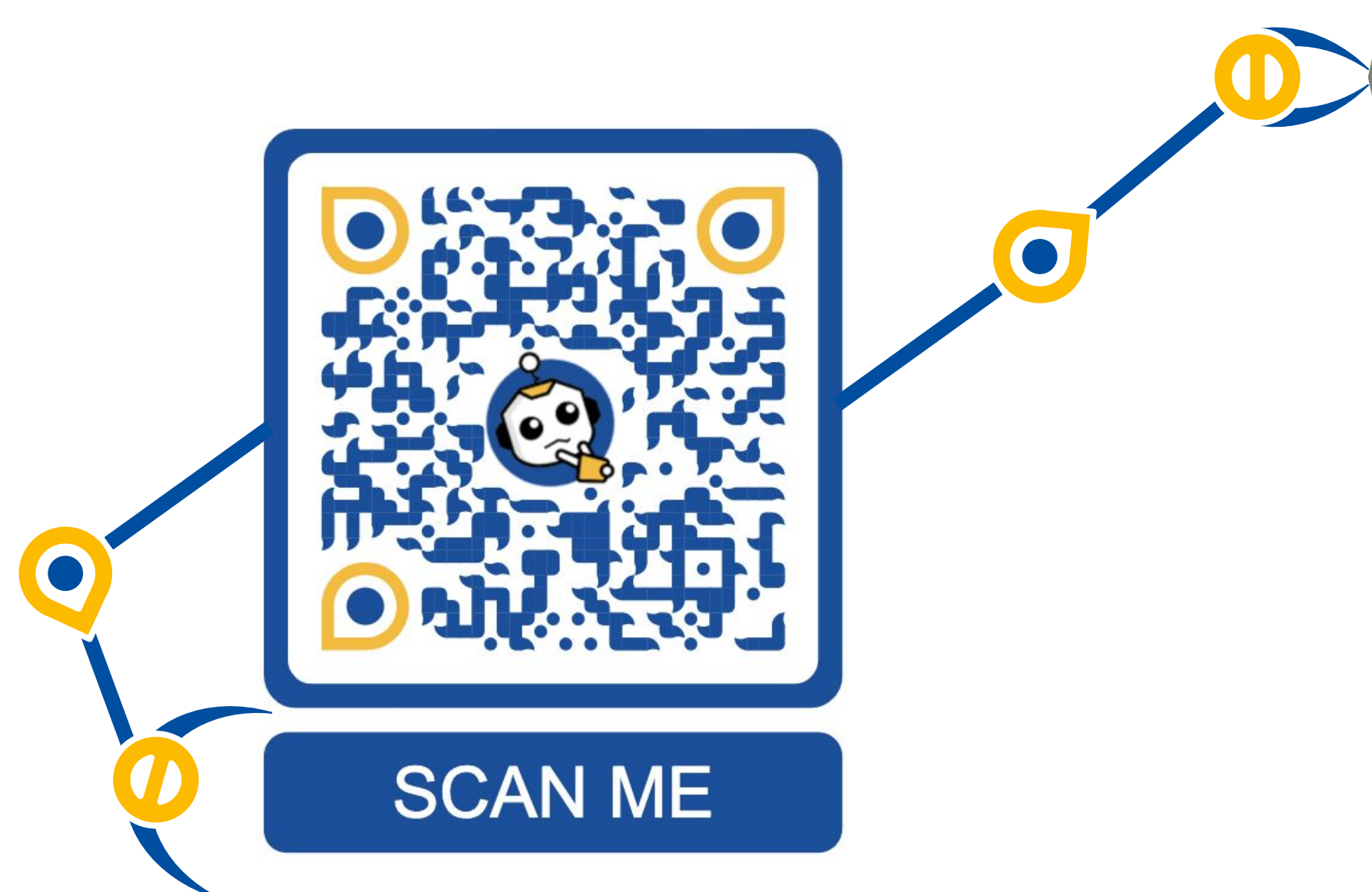
        mug, branch =
select_by_sceneGraph(mugs, branches);

```

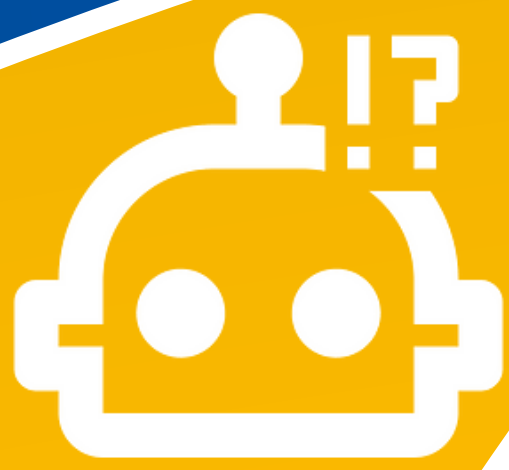
Attributes	Relations
 0: light blue cup on wooden table 1: pink cup on white napkin 2: rightmost wooden stick part of easel 3: middle wooden stick part of easel 4: left wooden stick part of easel 5: leftmost wooden stick part of easel 6: wooden easel stand supporting stick	0 to 2: attached to 2 to 0: supporting

Result

- Scene graphs capture useful object relationships (e.g., attached to, supporting)
- Ambiguous instructions are supported by an explicit relational explanation
- Single-image input limits depth reasoning (e.g., selecting the closest target)
- LMP's requiring hardcoded methods for selection restricts parameter breadth
- Flawed codebase impedes collection of numerical results and deeper evaluation

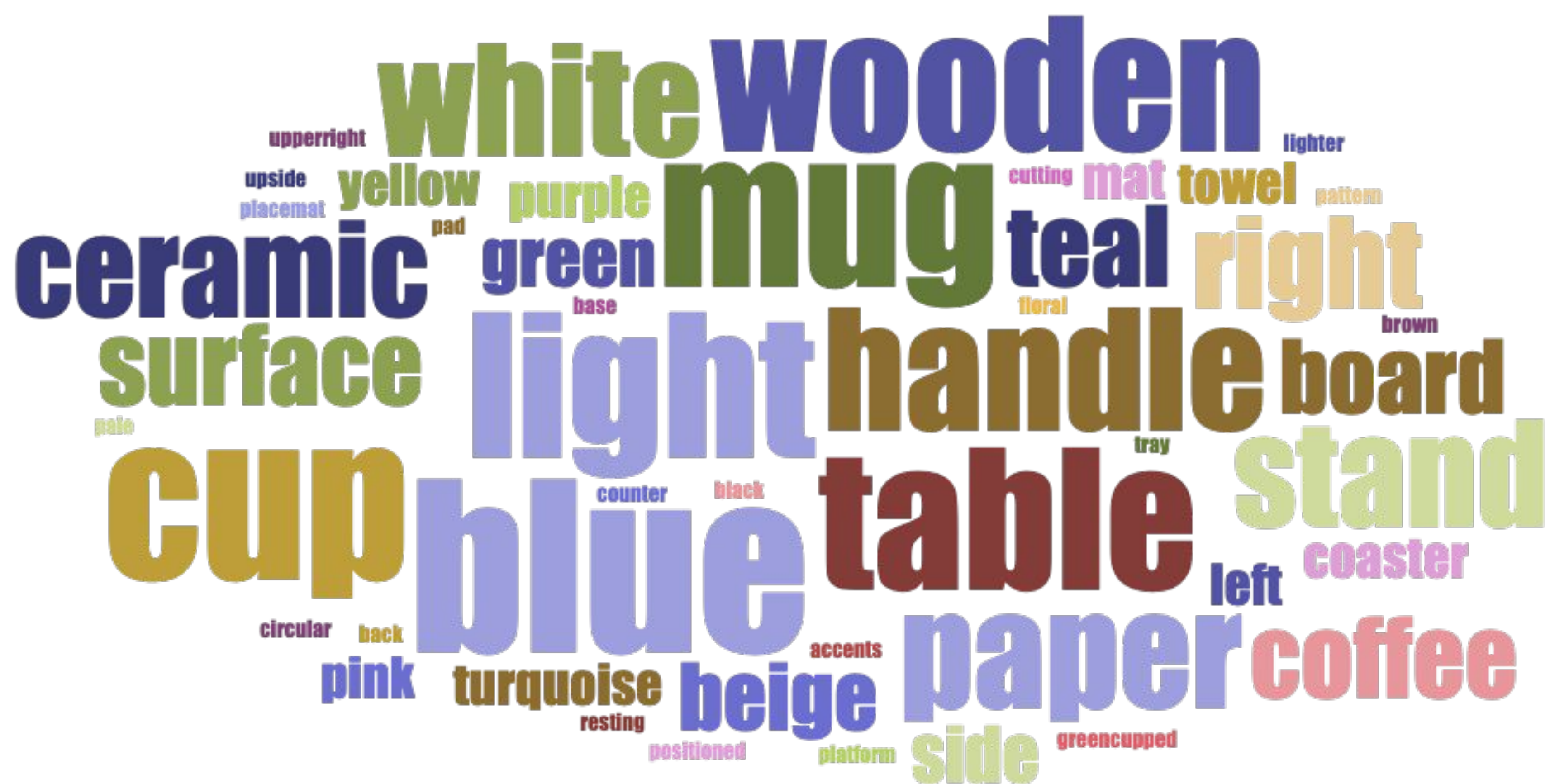
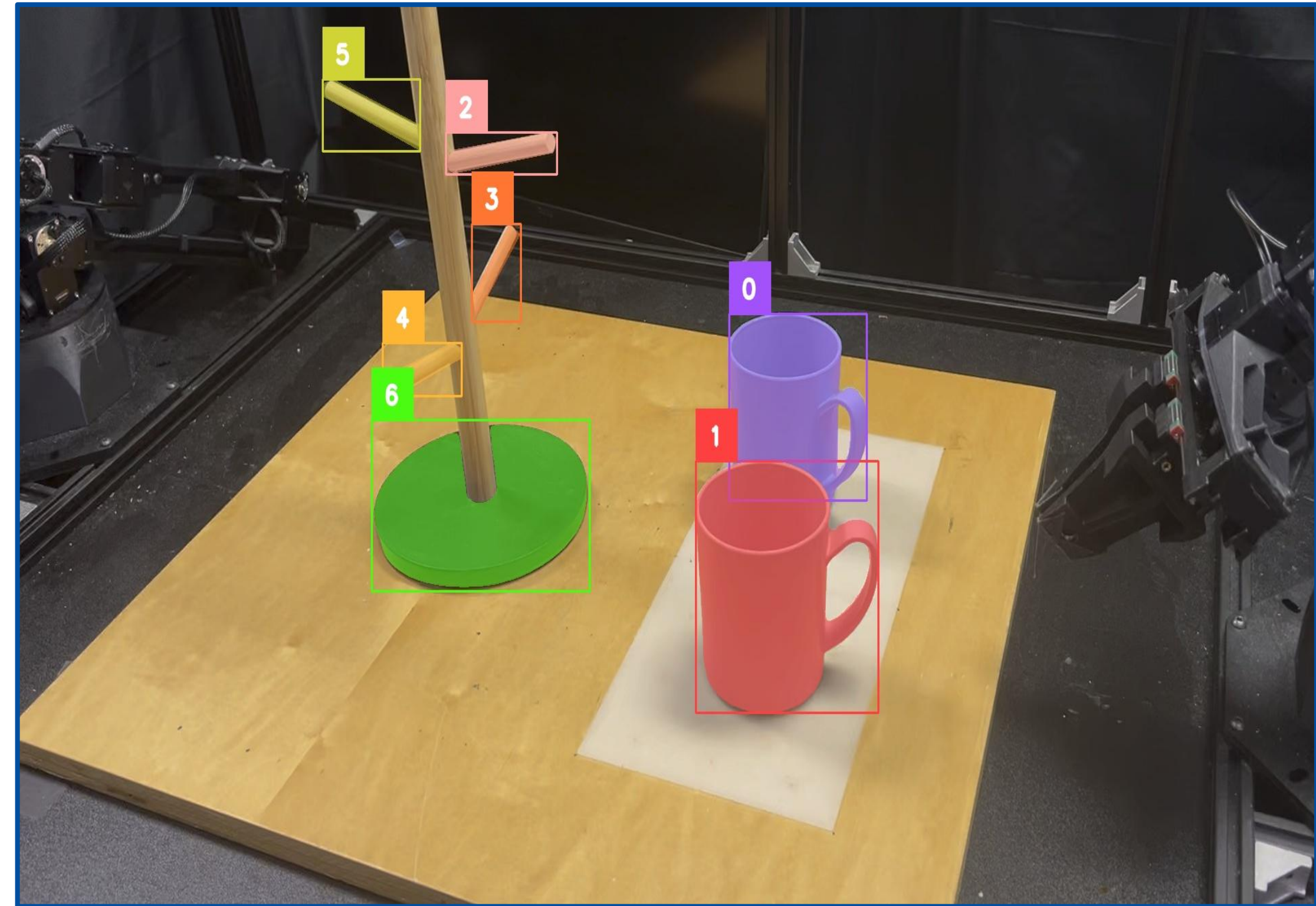


Scene Graphs provide in-depth external look-up data for VLM's



Mug 0

Branch 2



attached to

supporting

